

United States Telecom Services: Key Takeaways from Regional Fiber Connect

We attended the Fiber Broadband Association's Regional Fiber Connect conference in Boston. AI was the key theme both as a catalyst for fiber demand and as a driver of edge compute demand. Edge data centers were also highlighted as a potential way to get compute faster amid community pushback and longer construction timelines for HPC data centers.

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Fiber unit economics can support multiple passings. The Fiber Broadband Association estimates a TAM for fiber to the home (FTTH) of 130mn homes passed, accounting for multiple passings. The organization has seen more 2nd/3rd/4th fiber passings on a single home. FBA highlighted AI as a catalyst for fiber demand, as 75% of fiber users use AI daily compared to 60% of cable/coax users, 30% of FWA users, 20% of DSL users, and 20% of satellite users. Telehealth was also discussed as a potential demand driver as outpatient monitoring solutions require continuous connectivity and low latency.

Early signs of edge compute demand. CTC Technology and Energy, a communications, technology, and energy consultancy, is starting to see significant green shoots of edge compute demand as investors seek facilities for cost effective compute. They noted that AI adoption does not necessarily predicate edge compute, citing underlying growth in data use and privacy concerns, although it does strengthen the demand. Still, the company recognized there are questions around if there will be sufficient need for on-prem edge compute and on the ability for devices to support edge compute.

Fiber supply constraints and permitting are challenges to deployment. Fiber supply and permitting/regulation were both highlighted as challenges for fiber deployment. Separately, within data center interconnect, hyperscalers demand low latency and significant network capacity which adds additional challenges to deployment. Two innovations were highlighted: (1) hollow-core fiber which enables 30-35% lower latency than traditional glass-core fiber and (2) multi-core fiber which is useful as higher fiber strand fiber counts (e.g. ~13k+ strands) are not bendable and thus difficult to use to connect points.

Price and network are top drivers of customer switching. In general, Recon Analytics estimates that about half of home internet churn is related to customers moving addresses. They also noted that while price is the number one reason cited for switching providers, price is rarely the sole driver for switching. There is also a clear divergence within modalities: 40% of FWA customers say network performance

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is the top reason for leaving and 25% cite price while 29% of cable customers cite network performance and 38% cite price.

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